Computer Vision 24 Final

Faculty of Computer & Information Sciences
Ain Shams University
Course Code: SCO 421
Course Name: Computer Vision
Exam: (Final) 22/1/2024
Year: (1st term) 3rd & 4th Levels (SC-CSys)

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Offering Dept: Scientific Computing
Academic year: 2023-2024
Duration: 2 Hours
Total Grade: 50 Marks

Question (1) **Marks: 22**

A) [9 marks] State whether each of the following statements is True of False.

1. For symmetric filters, there is no difference between correlation and convolution. ( ) True
2. An edge is a place of rapid change in the image intensity function. ( ) True
3. The Harris detector is invariant to location but dependent on rotation and scale. ( ) False
4. The first derivative of the image intensity function has a peak at the edge. ( ) True
5. Consider an image of size $m \times n$ and a filter of size $k \times k$, the number of multiplications that are performed during this filtering operation is $m x n x k^2$. ( ) True
6. The Laplacian operator is useful in calculating the direction of an edge. ( ) False
7. It is cheaper to implement the Laplacian operator than the gradient operator. ( ) True
8. Normalizing the SIFT descriptors in the direction of the dominant gradient orientation would allow the descriptors to be invariant to image scaling. ( ) False
9. RANSAC is used to get rid of outliers in situations where there are wrong point correspondences. ( ) True

3- Harris detector is invariant to location “translation” and rotation but not scaling.

6- Laplacian filter provides only location of the edge, but the Gradient filter provides location, magnitude and direction of the edge.

8- Normalizing SIFT descriptors in the direction of dominant gradient orientation makes them invariant to viewpoint and illumination changes. Scale invariance in SIFT comes from its scale-space pyramid approach.

B) [13 marks] Answer the following questions in ONE word/expression:

1. What is the process that results from subtracting two blurred images with different blurring degrees from each other? Difference of Gaussians (DoG)
2. What is the indication of finding an edge in an image using Laplacian Filter? Zero-Crossing
3. Given the moment matrix of Harris at some image pixel as $\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$, what indicates that this pixel lies in a flat region? λ_1 and $\lambda_2 \approx 0$
4. What linear mathematical property a mask/kernel must have to be separable? Convolution
5. What kind of images that needs to be preprocessed using Gaussian filters? Noisy image
6. In order to apply instance segmentation, what task needs to be applied first to a given image? Object Detection
7. Which process/step does the FCN network for semantic segmentation make use of, to improve the segmentation accuracy? Skip Connections
8. What kind of a loss function a CNN-based network can use in order to verify that two images are similar or different? Triplet Loss
9. Which algorithm for region proposals does the Fast RCNN network of object detection use? Selective search
10. How many keypoint correspondences are needed to solve the Homography transformation problem, to find if two image pairs are scaled? 4
11. What is the main function that Bag Of Words (BOW) is used for, in the context of the image recognition task? Simplifying object representation / Classifying
12. What function does the U-Net model for semantic segmentation uses for the up-sampling path of the network? Skip connections
13. Which parameter(s) in the YOLO algorithm that determine(s) the size of the detected objects? Anchor Boxes

**Not Sure about 6, 11, 12 **

A) **[4 marks]** Given the 4x4 image below (**Figure a**), apply the median filter on the image pixels using the filter mask given in (**Figure b**) to calculate the new 4x4 image pixels. Consider the padding pixels as zeros.

100	110	160	120
80	90	100	100
90	90	50	120
40	100	19	150

Figure a

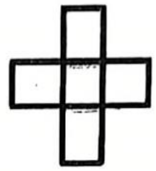


Figure b

Step1: Zero Padding

100	110	160	120
80	90	100	100
90	90	50	120
40	100	19	150

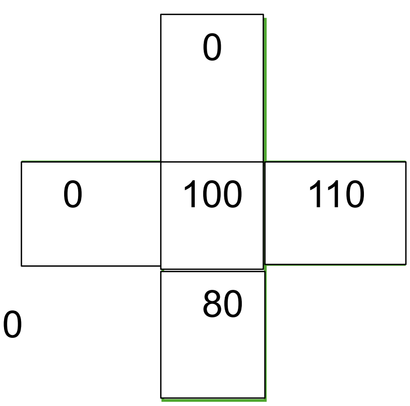
→

0	0	0	0	0	0
0	100	110	160	120	0
0	80	90	100	100	0
0	90	90	50	120	0
0	40	100	19	150	0
0	0	0	0	0	0

Step2: Apply Median

For each cell in original image, we apply median filter and assign in new image matrix.

For cell (1,1): 100



Values= 0, 0, 80, 100, 110

Median = 80

So, we'll replace 100 with 80 in the new image.

For cell (1,2): 110

	0	
100	110	160
	90	

Values= 0, 90, 100, 110, 160

Median = 100

So, we'll replace 110 with 100 in the new image.

The Final result:

80	100	110	100
90	90	100	100
80	90	90	100
40	40	50	19

B) [8 marks] Given the following filters, mention for each; the title of the filter and the function it does when convolved with an image. Use the provided Answer Table in your sheet.

$$\begin{matrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{matrix}$$

(a)

$$\begin{matrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{matrix}$$

(b)

$$\begin{matrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{matrix}$$

(c)

$$\begin{matrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{matrix}$$

(d)

$$\begin{matrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}$$

(e)

$$\begin{matrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{matrix}$$

(f)

$$\begin{matrix} -1/9 & -1/9 & -1/9 \\ -1/9 & 3/9 & -1/9 \\ -1/9 & -1/9 & -1/9 \end{matrix}$$

(g)

$$\begin{matrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{matrix}$$

(h)

Filter	Title	Function
a	Sobel-x	Detect vertical edges
b	Box Blur	Blur the image by averaging each pixel with its neighbors
c	Laplacian filter	Detecting edges zero-crossing based
d	Sobel-y	Detect horizontal edges
e	Shifting	Shift right by 1 pixel
f	Impulse filter	Return the original image unchanged
g	Laplacian filter normalized	Detecting edges normalized
h	Gaussian smoothing	Remove the noise from the image by blurring

Question (3) **Marks: 16**

A) [4 marks] The problem of content-based image retrieval (CBIR) is of significant interest, since Google and other search engines attempt to produce relevant images based on search criteria. **Answer the following questions:**

i. What makes image searching fundamentally more difficult than searching text?

ii. If you have two related problems:

1. Searching for similar images (e.g., all sunset pictures), or
2. Searching for matching images (e.g., rotated, resized versions of the same original image).

You would certainly want to extract different image features for those two problems. <u>**Suggest**</u> a model that is able to solve the CBIR task and allows to select the required features according to the searching problem.

I.

The Image searching fundamentally more difficult because of:

Semantic Gap: which is the difference between low-level features like (pixel values, edges, textures) that a computer can extract and high-level semantic meanings that humans associate with images.

II.

CNN Network for feature extraction, we can use improved version with skip connections and matching algorithms to get similar images.

B) **[12 marks]** Given an image of a global scene (**Image A**) and an image of an object (**Image B**), you are required to design an algorithm that is able to answer whether **Image A** contains **Image B** or not. In order to find a solution for this task, your algorithm should include two main procedures: First, a CNN-based object proposal procedure, that is able to take any image (e.g. **Image A**) and output the bounding boxes of all possible/candidate objects that may exist, irrespective of their category (**Image C**).

Second, a matching procedure, that based on color histograms (**Image D**), tells if **Image B** is considered a close match of any possible object found in **Image C**.

i. **Sketch** a general architecture of the CNN-based object proposal network. **Name** each part of this architecture. **Explain** in brief:
   1. The expected output of this network.
   2. The Ground Truth data that this network should be trained on.

ii. **Explain** in algorithmic steps the matching procedure, that takes as input **Image B** along with a sequence of the bounding boxes output of the object localization model. **Show** any preprocessing that needs to be done for such input if necessary. The final expected output of the matching procedure should be (Yes/No) indicating whether **Image B** exists in **Image A**, or not.

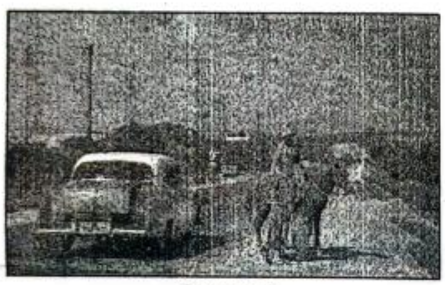


Image A

Image B

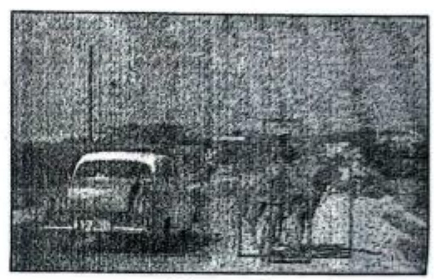


Image C

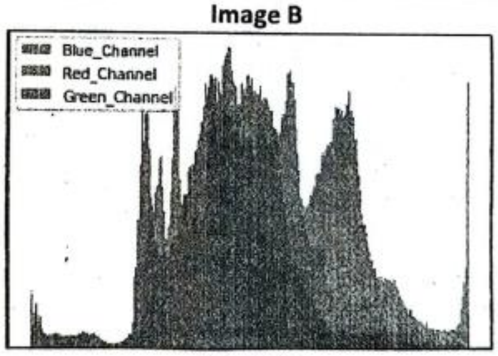


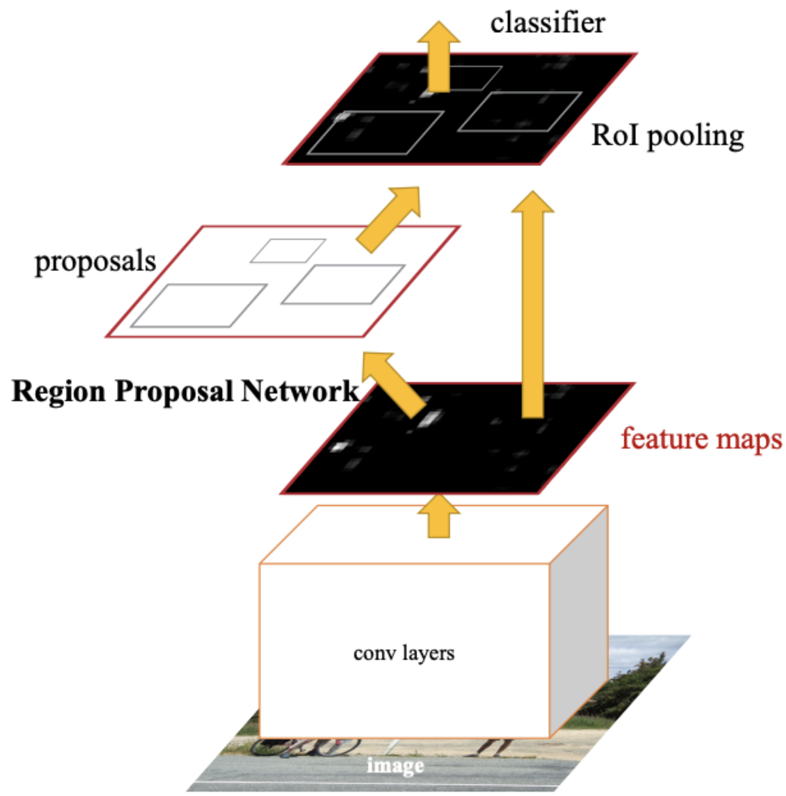
Image D

A color histogram models an image as a distribution of the image pixels color (R,G,B components of each pixel)

I.

1- Expected Output:

- Class Predictions
- Bounding Box Coordinates



2- Ground Truth Data:

- Bounding Box Coordinates
- Object Class Labels
- Background Regions

II.

Algorithmic Steps for Matching Procedure

1. **Preprocessing**:
    - preprocess the image (e.g., resizing, normalization) to match the input.
2. **Extract features from the object in the image**

3. **Bounding Box Evaluation:**
    - For each bounding box in the sequence:
    - Crop the image using the bounding box coordinates.
    - Extract features from the cropped image.

4. **Feature Comparison**:
    - Compare the features of the original object (from the full image) with the features extracted from each cropped bounding box. This can be done using Cosine similarity, Euclidean distance

5. **Thresholding**:
    - Set a threshold for similarity. If the similarity score is above the threshold, consider the images as a match.

6. **Final Decision**:
    - If any of the bounding boxes from Image A matches with the corresponding region in Image B, output "Yes".
    - If none of the bounding boxes match, output "No".